

# Generation Interconnection Request System Impact Study Report

## For Queue Project Q24-2426

Roanoke Rapids 138 kV  
July 30, 2025

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2426. The Interconnection Customer proposes a 80.0 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on June 28, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 464 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$62,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility          | Contingency                                | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------|--------------------------------------------|-------------|--------------|----------|
| Dpc Elroy-Garden 138kV      | Mccartney-N Tower Tap 345kV                | 105.8       | 551          | 6.7      |
| N Tower Tap-Dpc Elroy 138kV | Mccartney-North Tower Tap Substation 345kV | 101.8       | 853          | 6.0      |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                    | Required Upgrade           | Cost Estimate | Allocation to Project |
|-----------------------------|----------------------------|---------------|-----------------------|
| Dpc Elroy-Garden 138kV      | Reconductor line           | \$3,403,000   | 1% (\$39,000)         |
| N Tower Tap-Dpc Elroy 138kV | Reconductor line           | \$2,270,000   | 1% (\$23,000)         |
|                             | Total allocated to project |               | \$62,000              |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.